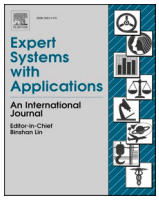
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Modelling long medical documents and code associations for explainable automatic ICD coding

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ABSTRACT

Quick and accurate International Classification of Diseases (ICD) code assignment is vital for billing, reimbursement and medical research. Owing to the labour-intensive and error-prone nature of manual coding, automatic ICD coding using deep learning methods has flourished. However, this task is still challenging because of (1) the interpretability of coding, (2) lengthy clinical documents and (3) long-tail label distribution. In the current study, we propose a novel automatic ICD coding framework to address these issues. First, a biomedical-specific pre-trained language model, Clinical-Longformer, is used as an encoder, which generates meaningful representations of long clinical documents by injecting rich medical knowledge and capturing long-distance dependence among tokens. Second, a decoding architecture that combines the multi-synonym attention mechanism, hierarchical curriculum learning and distribution-balance loss is designed to perform ICD code prediction. The decoder improves the tail-end performance by fully capturing code associations in terms of semantics, structure and co-occurrence. In addition, the label-wise attention mechanism provides the interpretability of prediction. Experimental results on benchmark MIMIC-III datasets indicate that our model achieves higher F1 scores than previous state-of-the-art baselines. Our suggested model is expected to serve as an aid in improving the efficiency of manual ICD coding and to offer insights for other long text classification tasks with multiple label associations.

1. Introduction

In the healthcare system, International Classification of Diseases (ICD) codes are designed to standardise diagnoses and procedures during patients' hospitalisation, which is helpful for various clinical tasks, including billing, reimbursement, health statistics and medical research. Traditionally, ICD coding tasks are carried out by specialised coders based on electronic medical records (EMRs) (Kaur et al., 2022). Excessive medical records, irregular physician writing and complex coding rules make manual coding labour-intensive and error-prone. The automatic ICD coding system can improve coding efficiency and has attracted increasing attention from scholars since the 1990 s (Dong et al., 2022).

Typically, automatic ICD coding is treated as a multi-label text classification issue, i.e. assigning appropriate ICD codes to specific clinical documents (Teng et al., 2022). With the successful application of neural networks in natural language processing (NLP) tasks, research on automatic ICD coding using deep learning algorithms has flourished. Among the algorithms, convolutional attention for multi-label (CAML) classification proposed by Mullenbach et al. (2018) is a landmark model that applies a label-wise attention mechanism to extract relevant clinical text snippets for each code, achieving the interpretability of medical decisions. The label-wise attention mechanism relies on the effective interaction between text representations and codes, which suffer from two challenges. (1) The clinical text is noisy and lengthy, making it difficult to extract meaningful document representation (Yu et al., 2019). (2) The ICD code space is vast and follows a long-tail distribution, resulting in an arduous task to learn the features of infrequent codes (Blanco et al., 2019). To address these two issues, many subsequent studies have improved on CAML.

For the representation learning of clinical documents, most of the existing studies developed various recurrent neural networks (RNNs) or convolutional neural networks (CNNs) configurations. The RNN-based models (Baumel et al., 2018; Dong et al., 2021; Shi et al., 2017; Sun

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et al., 2022; Vu et al., 2020) can mine sequence relationships of clinical inputs, which adopted the Bi-directional Long Short-Term Memory (Bi-LSTM) and Bi-directional Gated Recurrent Unit (Bi-GRU) networks to model clinical documents. The CNN-based models allow for the extraction of the local spatial structure of medical texts. Relevant researchers have proposed many variants of CNNs, such as multi-channel CNNs (Mayya et al., 2021), dilated CNNs (Cao et al., 2020b), multi-filter CNNs (Hu et al., 2021; Li & Yu, 2020) and residual CNNs (Liu et al., 2021), to obtain better document representation. However, CNNs and RNNs are still feeble at encoding long sequences of clinical documents. Recently, motivated by the advantages of Transformer architecture that can capture long-distance dependencies among tokens, several biomedical-specific pre-trained language models (PLMs), including BioBERT (Lee et al., 2020), ClinicalBERT (Alsentzer et al., 2019), PubMedBERT (Gu et al., 2021) and Clinical XLNet (Huang et al., 2020), have been proposed to enhance medical text representation. A few studies (Biswas et al., 2021; Gao et al., 2021; Huang et al., 2022; Pascual et al., 2021) have embedded these PLMs into ICD coding tasks and adopted text truncation, chunking or selection strategies to shorten the input sequence to meet the maximum sequence length limitation (512 tokens). However, these text processing strategies usually lead to information loss, resulting in poorer performance than RNN/CNN-based models. Lately, Clinical-Longformer and Clinical-BigBird, two long-sequence biomedical-specific PLMs that can process up to 4,096 tokens of clinical text at once, have been proposed (Li et al., 2022). Although Clinical-Longformer has achieved the best performance among all PLMs in 10 medical NLP tasks, it has not yet been applied to automatic ICD coding tasks. In the present study, we try to explore whether it can enhance the efficiency of ICD coding through its superior ability to encode long clinical documents.

To improve the tail-end performance, many studies used balanced loss functions, such as Focal Loss (FL) (Lin et al., 2017) and Asymmetric Loss (ASL) (Ridnik et al., 2021), for cost-sensitive learning. Recent studies focus on better modelling rare codes by capturing their associations with frequent codes. On the one hand, some studies exploited code descriptions to extract the semantic relevance of codes. For example, Kim & Ganapathi (2021) and Yuan et al. (2022) used CNNs and Bi-GRU to encode code and synonym descriptions for initialising the query vector in the label-wise attention layers. On the other hand, many researchers leveraged the ICD taxonomy's hierarchical architecture to capture the structural dependencies of codes. Rios & Kavuluru (2018) and Ren et al. (2022) adopted graph convolutional neural networks (GCNNs) or hierarchical learning mechanism to train ICD code embedding, which uses the mutual exclusion and inheritance relationship between sibling and parent–child ICD nodes. However, the semantic relevance and structural dependencies of ICD codes rely on third-party knowledge bases, which only reflect code association statistically. Code co-occurrence in dynamic data also reflects the intrinsic association of codes, which can provide additional knowledge for code prediction. For instance, Cao et al. (2020a) and Wu et al. (2022) mined code co-occurrence patterns to predict rare codes by constructing code co-occurrence graphs and learning code probability distributions. The above studies have demonstrated the critical role of code semantics, code structure and code co-occurrence in improving tail-end performance but failed to use all three elements together. In the current study, we wish to establish a framework that could simultaneously extract code associations in these three aspects to help the feature learning of rare codes.

In short, we develop a novel automatic ICD coding model that focuses on enhancing long medical document representation and capturing various code associations. The proposed framework consists of four parts: (1) long-sequence biomedical-specific PLMs for medical document representation; (2) multi-synonym attention mechanism for code semantic relevance; (3) hierarchical curriculum learning for code structure dependency and (4) distribution-balanced loss (DBL) (Wu et al., 2020) for code co-occurrence pattern. Experiments on the MIMIC-III dataset show the superior performance of our model. For the top 50 most frequent ICD-9 codes, our model achieves macro-F1 and micro-F1 of 0.688 and 0.727, respectively, outperforming previous state-of-the-art (SOTA) methods. For the full ICD-9 codes, our decoder improves the performance of RNN/CNN-based models on all AUC, F1 and precision metrics. The suggested framework is expected to be an auxiliary tool to help coders assign the correct ICD codes to EMRs. The main contributions of this study are 3-fold:

(1) This is the first study to introduce Clinical-Longformer into automatic ICD coding tasks, which extracts meaningful clinical document representations by capturing long-distance dependence among tokens and reduces information loss in short-sequence PLMs. The results provide a reference for using PLMs in automatic ICD coding tasks.
(2) We propose a novel code association learning architecture, which combines multi-synonym attention mechanism, hierarchical curriculum learning, and DBL to simultaneously capture the semantic relevance, structural dependency, and co-occurrence patterns of ICD codes. This architecture improves tail-end performance of ICD coding tasks and can offer guidance for other multi-label classification tasks with label associations.
(3) The experimental results and visual analysis on the MIMIC-III dataset demonstrate the interpretability of the proposed ICD coding framework and its superiority in modeling long clinical documents and code correlations, which provides substantial assistance for manual coders' work.

The current study is divided into six sections. Section 2 reviews the related studies on automatic ICD coding. Section 3 describes the details of the suggested model. Section 4 conducts comparison experiments and result analysis. Section 5 discusses the effects of several components in our model. Finally, Section 6 provides the conclusion and future directions.

2. Related studies

This section first analyses the challenges of automatic ICD coding tasks and then details some efforts to address these challenges.

2.1. Automatic ICD coding

The automatic ICD coding task is to assign appropriate ICD codes to a given clinical document in an automated manner. Existing studies on automatic ICD coding generally fall into the following two categories.

Rule-based approaches: Mining coding rules from ICD coding guidelines and converting them into executable if-else logical programmes (Farkas & Szarvas, 2008). Such systems rely on medical experts' experience and are less flexible or portable. Moreover, as the number of codes increases, extending the rules is extremely time-consuming and conflicts between various rules are likely to occur.

Learning-based approaches: Treat the coding task as a multi-label text classification issue (Zhang et al., 2022), learning underlying coding patterns from historical data without over-reliance on medical experts. Previous studies (Atutxa et al., 2017; Xu et al., 2019) extracted features of clinical texts by bag-of-words or term frequency-inverse document frequency methods and then constructed code classification models using traditional machine learning (ML) algorithms. Recently, deep learning–based models have designed different neural networks to learn the mapping between EMRs and ICD codes, which have outperformed ML systems but still face the following challenges (Huang et al., 2019).

(1) **The interpretability of coding:** Providing a reasonable interpretation for predicted code is critical in clinical practice, yet end-to-end deep neural networks suffer from black box problems (Moradi & Samwald, 2021).

(2) **The lengthy clinical documents:** The clinical text may be lengthy (average 1,500 tokens in the MIMIC-III dataset). Extracting crucial information from clinical documents and learning meaningful representations for ICD code prediction is extremely difficult (Wang et al., 2020).
(3) **The unbalanced code distribution:** The ICD code space is extremely huge (with over 15,000 and 60,000 codes in ICD-9-CM and ICD-10-CM taxonomies) and follows a long-tailed distribution (with $>$ 90 % of the top 50 codes). For infrequent codes, only a few instances are available for training, which affects the feature learning of rare diseases and reduces coding accuracy (Song et al., 2021).

To address the three challenges mentioned above, scholars have performed many studies.

2.2. Interpretability

In deep learning models, the attention mechanism is the most widespread approach for interpretability. For each label, the label-wise attention mechanism allows assigning specific weights to different text snippets, thus making predictions interpretable. About automatic ICD coding, Mullenbach et al. (2018) used CNNs to learn the document representation, followed by a label-wise attention mechanism to extract the most relevant tokens for prediction, providing a word-level explanation for each label. Baumel et al. (2018) and Dong et al. (2021) suggested a hierarchical attention gated recurrent unit (HA-GRU) and a hierarchical label–wise attention network (HLAN) to predict ICD codes, where a sentence-level attention mechanism produced sentence-level explanation.

2.3. Lengthy document representation

For the semantic representation of clinical documents, scholars usually use CNN, RNN or Transformer-based models to mine document features.

RNN-based methods: RNNs and their variations, such as LSTM and GRU, can extract sequential information from variable-length inputs through memory mechanisms and have been introduced into automatic ICD coding tasks. For example, Shi et al. (2017) proposed a hierarchical label-wise attention LSTM architecture that uses LSTM to encode clinical documents and code descriptions. The Bi-LSTM network (Vu et al., 2020) and Bi-GRU network (Baumel et al., 2018; Dong et al., 2021) have also been adopted to learn the representations of clinical text. Subsequently, Sun et al. (2022) added a recalibrated attention module following the Bi-GRU network, where the cascaded convolutional structure can better handle lengthy documents.

CNN-based methods: CNNs can extract key n-gram features of sentences and encode local semantic information efficiently. For long clinical texts, the simplicity and parallel processing friendliness of CNNs make them more attractive for automatic ICD coding. The most classic models are CAML and description regularised (Mullenbach et al., 2018), where a single-channel CNN layer is adopted to encode clinical documents. Mayya et al. (2021) presented the enhanced CAML (EnCAML), which captures various information through multi-channel CNNs. Subsequently, dilated CNNs (Cao et al., 2020b), multi-filter CNNs (Hu et al., 2021) and multi-filter residual CNNs (MultiResCNN) (Li & Yu, 2020) have been introduced to process clinical text fragments with different lengths and reduce information loss.

Transformer-based methods: Owing to the excellent encoding capabilities, Transformer architecture has become another paradigm for NLP tasks. TransICD (Biswas et al., 2021) is a typical example that performs automatic ICD coding with the Transformer encoder. Since 2019, the emergence of biomedical-specific PLMs, such as BioBERT (Lee et al., 2020), ClinicalBERT (Alsentzer et al., 2019), PubMedBERT (Gu et al., 2021) and Clinical XLNet (Huang et al., 2019), has provided strong support for medical text encoding. However, they have a maximum sequence length limitation of 512 tokens and cannot be applied to the ICD coding task directly. For this issue, Pascual et al. (2021) proposed a BERT-ICD model, which explores five text truncation strategies to shorten the lengthy clinical text to 512 tokens for encoding but results in severe information loss. Feucht et al. (2021) proved that the Longformer, which can handle longer sequences of up to 4,096 tokens, has helped the description-based label attention classifier (DLAC) perform better in ICD coding tasks. However, the performance is still worse than CNN/RNN-based SOTA models since Longformer is pre-trained on Wikipedia and BookCorpus rather than on the medical corpus. Recently, Li et al. (2022) introduced Clinical-Longformer, which is pre-trained on the large-scale medical corpus and significantly outperforms various short-sequence PLMs in 10 medical NLP tasks. Unfortunately, it has not yet been used in automatic ICD coding tasks.

Generally, compared to RNN/CNN-based models, the Transformer-based models have superior semantic encoding capability, but the limitation of sequence length significantly weakens their performance in ICD coding tasks. We will introduce the recently presented Clinical-Longformer for ICD coding, as it offers excellent access for representation learning of lengthy clinical documents.

2.4. Long-tail label classification

In multi-label classification tasks, balancing loss functions improved on the binary cross-entropy (BCE) loss, such as FL (Lin et al., 2017), ASL (Ridnik et al., 2021) and DBL (Wu et al., 2020), provide enhancements to address the label imbalance problems. The modulating factor of ASL and FL assigns lower weights to high-frequency labels, thus enhancing the learning of low-frequency labels. By integrating rebalancing weighting and negative tolerance regularisation, the DBL captures the label co-occurrence information that is critical in multi-label scenarios. For automatic ICD coding tasks, ASL and FL have been introduced in (Biswas et al., 2021; Liu et al., 2021; Sun et al., 2022) to guide the learning process. However, the DBL, which has shown good performance in computer vision, has not yet been applied to ICD coding tasks. Actually, researchers have focused on capturing code associations to mitigate the deficiency caused by long-tail. Related studies can be divided into the following three groups.

Code semantic: The characteristics of rare diseases should be similar to those of diseases with analogous text descriptions. Considering this, Dong et al. (2021) and Mayya et al. (2021) encoded ICD code descriptions and embedded them in the attention mechanism, which exploits the semantic correlation between high- and low-frequency codes to make better predictions. Yuan et al. (2022) further introduced code synonym descriptions in the Unified Medical Language System (UMLS) to help code representation learning and proposed a Multiple Synonyms Matching Network (MSMN) for better code classification.

Code hierarchy: ICD codes are arranged in a tree-like form, where parent nodes denote generic disease categories and child nodes indicate more specific diseases, implying exclusive and inherited relationships among codes. To exploit these relationships, Vu et al. (2020) designed a hierarchical joint learning mechanism (JointLAAT) for rare ICD code prediction. Cao et al. (2020a) proposed a hyperbolic and co-graph representation method (Hypercore), which embeds ICD codes into hyperbolic space to capture their structure associations.

Code co-occurrence: Since some diseases are concurrent or causally related, they usually co-occur in clinical records. For example, ICD 9 code 518.81 (acute respiratory failure) is a complication of 486 (pneumonia), which occurred 1,500 times simultaneously in the MIMIC-III dataset. Learning code co-occurrence patterns in the data can provide additional knowledge to predict rare codes correctly. For this reason, Cao et al. (2020a) constructed a code co-occurrence graph to capture code co-occurrence. Wu et al. (2022) introduced the joint attention networks (JAN), in which CorNet blocks take advantage of label co-occurrence.

In summary, code descriptions, code structure and code co-occurrence all facilitate the performance of rare codes. However, existing studies for automatic ICD coding do not consider all three aspects simultaneously. In the present study, we combine multi-synonym attention mechanisms, hierarchical curriculum learning and DBL to improve tail-end performance by simultaneously capturing the semantic, structural and co-occurrence associations between codes.

Table 1 compares the relevant studies on automatic ICD coding, demonstrating the contribution of this study.

3. Methodology

3.1. Problem formulation

Suppose a given data set $\{x, y\}_{i=1}^{N}$ contains N instances, where $x \in R^n$ denotes a clinical document with n tokens, $y \in \{0,1\}^{|C|}$ is a $|C|$-dimensional binary vector representing the full set of ICD codes, and each bit y_i indicates whether the corresponding code is relevant to the document. The task is to train $|C|$ binary classifiers based on a training set and then predict a set of ICD codes for a given clinical document input.

3.2. Model architecture

This section details the architecture of the proposed model, which is shown in Fig. 1. In addition, we include a convenient summary of some crucial notations in Table 2.

3.2.1. Document representation module

The goal of this module is to extract meaningful medical document representations. Previous studies usually adopted various variants of RNN/CNN and short-sequence PLMs as the encoders. In the current study, we introduce Clinical-Longformer as the encoder, which contains richer medical knowledge than CNN/RNN and causes less information loss than short-sequence PLMs, thus generating more semantic document representation. Specifically, given a clinical document x, we first process it into token sequence $x = \{t_1, t_2, t_3, ..., t_n\}$ as input, where t_i is the vocabulary index of the i th token and n denotes the document length. Subsequently, each token t_i is converted into token embedding $h_i \in R^{d_h}$ via a pre-trained Clinical-Longformer and combined into the clinical document representation matrix $H = \{h_1, h_2, h_3, ..., h_n\} \in R^{n \times d_h}$, where d_h is the output size of the Clinical-Longformer model.

3.2.2. Code representation module

After obtaining the document representations, we need to match them with ICD codes according to the code representations. Unlike previous studies that learned code representations based on single code descriptions, this study introduces code synonyms from UMLS (Bodenreider, 2004) to expand code representations by capturing more comprehensive code expressions. The specific process is as follows.

First, align each code description with the Concept Unique Identifier in UMLS and select corresponding synonyms. For example, code 250.00 is described in the ICD as 'type II diabetes mellitus', and its synonyms in the UMLS include 'insulin-resistant diabetes', 'non-insulin dependent diabetes', 'DM2′ and 'T2DM'. Mathematically, for a code description l^1, its synonyms are denoted as $\left\{l^2, ..., l^M\right\}$, where the j th synonym l^j consists of the words $\left\{l_i^j\right\}_{i=1}^{n_j}$.

Then, we encode each synonym using the same method as (Yuan et al., 2022). Specifically, each token $\left\{l_i^j\right\}_{i=1}^{n_j}$ in l^j is mapped to $\left\{e_i^j\right\}_{i=1}^{n_j}$ by d_e-dimensional word embeddings generated by Word2Vec. Subsequently, the embedding matrix $E^j = \left\{e_1^j, e_2^j, ..., e_{n_j}^j\right\} \in R^{n_j \times d_e}$ is fed into an LSTM layer and a linear layer to acquire a context-specific representation matrix $H^j = \left\{h_1^j, h_2^j, ..., h_{n_j}^j\right\} \in R^{n_j \times d_h}$, where the dimension d_h is the same as the clinical document representation. Finally, using the max-pooling operation, the code synonym representations $\{q^j\}_{j=1}^{M}$ and q_l can be generated as follows:

$$q^j = MaxPool\left(h_1^j, h_2^j, ..., h_{n_j}^j\right) \in R^{d_h}, j = 1, 2, ..., M \quad (1)$$

Table 1
Relevant research on automatic ICD coding.

Method	Explainability	Document representation	Code			Literature
			Description	Structure	Co-occurrence	
RNN-based methods	Attention	LSTM	LSTM	×	×	AttentiveLSTM (Shi et al., 2017)
	Sentence-level attention	Bi-GRU	×	×	×	HA-GRU (Baumel et al., 2018)
	Attention	Bi-LSTM	×	Hierarchical joint learning	×	JointLAAT (Vu et al., 2020)
	Hierarchical attention	Bi-GRU	Bi-GRU	×	×	HLAN (Dong et al., 2021)
	Multi-synonyms attention	Bi-LSTM	Bi-LSTM	×	×	MSMN (Yuan et al., 2022)
CNN-based methods	Attention	CNNs	×	×	×	CAML (Mullenbach et al., 2018)
	Attention	CNNs	CNNs	×	×	DR-CAML (Mullenbach et al., 2018)
	Attention	Multi-channel CNNs	×	×	×	EnCAML (Mayya et al., 2021)
	Attention	Multi-filter residual CNNs	×	×	×	MultiResCNN (Li & Yu, 2020)
	Attention	CNNs	×	Hyperbolic	GCNNs	HyperCore (Cao et al., 2020a)
	Joint Attention	CNNs	CNNs	×	CorNet	JAN (Wu et al., 2022)
Transformer- based methods	×	Transformer	×	×	×	TransICD (Biswas et al., 2021)
	×	Text truncation + PubMedBERT.	×	×	×	BERT-ICD (Pascual et al., 2021)
	Attention	Longformer	Word2vec	×	×	DLAC (Feucht et al., 2021)
	Multi-synonyms attention	Clinical-Longformer	Clinical-Longformer	Hierarchical curriculum learning	Distribution balanced loss	This study

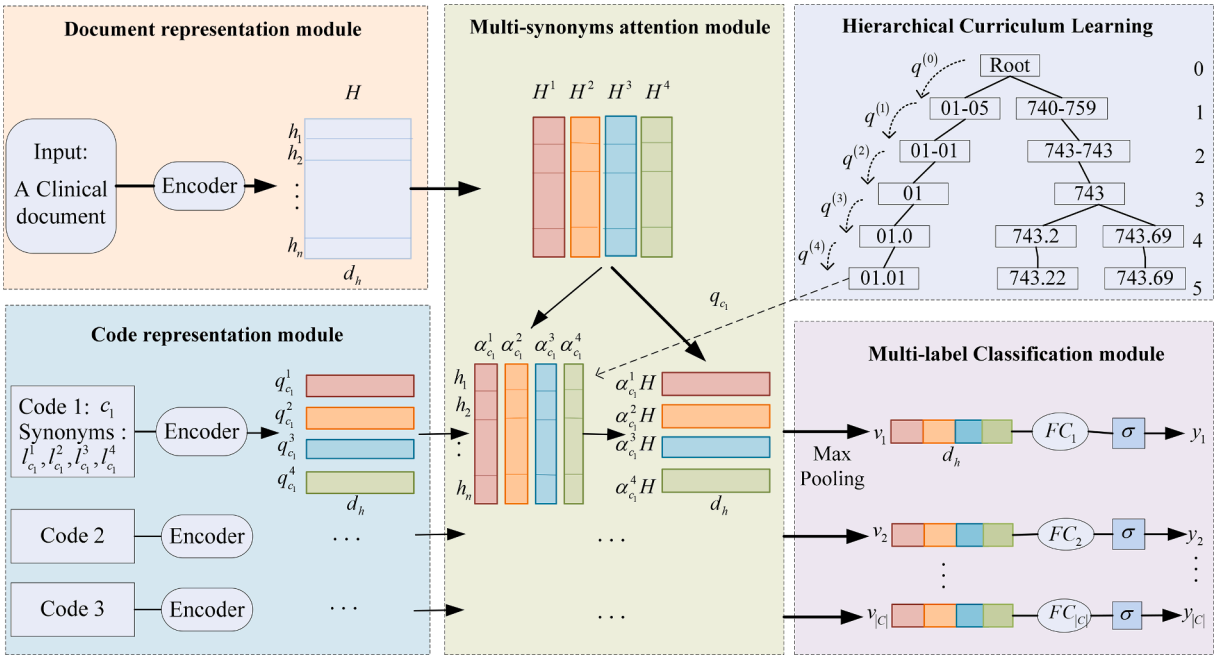


Fig. 1. Architecture of our model.

Table 2
Notation definition.

Symbol	Definition or description
N	The number of clinical documents
C	The number of codes
x	The clinical document description
n	The length of clinical document
h	The word vector of clinical document obtained by Clinical-Longformer
d_h	The dimension of word vector
l	The code synonym description
q	The code synonym vector obtained by Clinical-Longformer and max-pooling
M	The number of heads (i.e., the number of code synonyms)
α	The attention score of words in clinical document for each code
v	The code-specific clinical document vector obtained by attention mechanism and max-pooling
σ	The sigmoid activation function

$$q_l = MaxPool\left(q^1, q^2, ..., q^M\right) \in R^{d_h} \tag{2}$$

3.2.3. Multi-synonym attention module

After obtaining the token-wise document representations H and code synonym representations $q^j \in R^{d_h}$, we aim to convert H into code-specific representations via an attention mechanism. Unlike previous studies that only used label representation as an attention query, we adopted a multi-synonym attention mechanism to interact with document representations with multiple synonyms. This is inspired by multi-head attention (Vaswani et al., 2017) and can be formulated into the following steps.

First, the document representation matrix $H \in R^{n \times d_h}$ is divided into M heads $H = H^1, ..., H^M$, $H^j \in R^{n \times \frac{d_h}{M}}$, where M indicates the number of selected synonyms. Subsequently, by the linear transformations of H^j and the query matrix $\widehat{q}^j$, we can obtain attention scores $\alpha_l^j \in R^n$. Next, the document representations linked to code synonym l^j are calculated as $\alpha_l^j H \in R^{d_h}$, which will be aggregated to code-specific document representations by a max-pooling operation: $v_l \in R^{d_h}$. Finally, all v_l can be aggregated into the code-specific document representation matrix $V = \{v_1, v_2, ..., v_{|C|}\} \in R^{|C| \times d_h}$. The related equations are as follows:

$$\alpha_l^j = softmax\left(W_Q \widehat{q}^j \cdot \tanh\left(W_H H^j\right)\right) \in R^n \tag{3}$$

$$v_l = MaxPool\left(\alpha_l^1 H, ..., \alpha_l^M H\right) \in R^{d_h} \tag{4}$$

It should be emphasised that when initialising the query matrix $\widehat{q}^j$, we not only use the code synonym representation q^j but also consider the code hierarchy information, as described in Section 3.3.

3.2.4. Multi-label classification module

The ultimate goal is to construct $|C|$ binary classifiers that use label-specific document representation V for prediction. First, feed the V into a linear layer that projects the document representation dimension into the label space size. Subsequently, a sum-pooling operation is performed to generate the score vector $\widetilde{y} \in R^{|C|}$ of all ICD codes. The output probabilities $\widehat{y} \in R^{|C|}$ between 0 and 1 are obtained through a sigmoid activation function. Finally, ICD codes that exceed a predefined threshold t are assigned to the document. This process can be expressed as follows:

$$Y = VW_L + b_L \in R^{|C| \times |C|} \tag{5}$$

where W_L is the linear weight and b_L is the bias.

$$\widetilde{y} = SumPool(Y) \in R^{|C|}, where \widetilde{y}_i = \sum_{j=1}^{l} Y_{ij} \tag{6}$$

$$\widehat{y} = sigmoid(\widetilde{y}) \in R^{|C|} \tag{7}$$

3.3. Training

To train $|C|$ binary classifiers over the code set C, we introduce a hierarchical curriculum learning algorithm proposed in (Ren et al., 2022), which includes the following steps.

Step 1: Construct a five-level code tree (see Fig. 1), where each code $c \in C$ is located at the last level.

Step 2: Generate a hierarchical training curriculum to sequentially predict the labels at each level.

Step 3: Output the predicted probability of each code at the final level in the tree structure.

The core idea of the algorithm is that the trained parameters of

parent nodes can be used to initialise the parameters of child nodes when training per-label attention networks, thus achieving knowledge transfer. In the present study, we use this idea to train the model, but perform the following improvements.

(1) During the training process of the last level, we leverage parameters from the parent node and code synonym representation q^j to determine the query matrix $\widehat{q}^j$ in the multi-synonym attention module. Specifically, for code c_i (such as 250.00 'type II diabetes mellitus'), the query vector from the parent node (250.0 'diabetes mellitus without mention of complications') is used as the initial query vector q_{c_i}, and then the representation $q^j_{c_i}$ of code synonyms ($j = 1, 2, 3, 4, 5$ corresponds to 'type II diabetes mellitus', 'insulin-resistant diabetes', 'non-insulin dependent diabetes', 'DM2' and 'T2DM', respectively) is adopted to modify the query vector by Eq. (6). In this way, the medical knowledge contained in code hierarchy and code synonym descriptions is fully utilised.

$$\widehat{q}^j_{c_i} = q_{c_i} + fc\left(q^j_{c_i}\right) \tag{8}$$

where $\widehat{q}^j_{c_i}$ is the final query vector and $fc(\cdot)$ is the fully connected layer.

(2) In addition, the choice of the loss function that directs the learning of neural networks is crucial. To deal with label imbalance, some studies have selected FL and ASL to optimise the model. The studies allow the model to focus on identifying infrequent labels by down-weighting high-frequency labels. In the current study, we adopt DBL, which considers label co-occurrence and has been proven to be suitable for multi-label text classification with label imbalance and label dependence. Therefore, applying DBL to the automatic ICD coding task can take advantage of the co-occurrence relationship of codes when predicting low-frequency codes.

For each ICD code, the learning goal is to minimise DBL, which can be expressed as follows:

$$L_{DB} = -\sum_{i=1}^{|C|} y_i \widehat{r}_{DB}(1-\widehat{y}_i)^{\gamma}\log(\widehat{y}_i) + (1-y_i)\widehat{r}_{DB}\frac{1}{\lambda}(\widehat{y}_i)^{\gamma}\log(1-\widehat{y}_i) \tag{9}$$

where y_i is the ground truth label and $\widehat{y}_i$ is the prediction probability. Based on BCE loss, the DBL adds three components to rebalanced weighting. First, a modulating factor $(1-\widehat{y}_i)^{\gamma}$, where $\gamma \geqslant 0$, is used to assign higher weight to the 'hard-to-classify' instances. Second, the scale factor $\frac{1}{\lambda}$, where $\lambda \geqslant 1$, is introduced to assign a lower weight to 'easy-to-classify' instances. Third, $r_{DB} = P^C_i / P^I$ is normalised rebalanced weight and is mapped to $[\alpha, \alpha+1]$ with a smoothing function $\widehat{r}_{DB} = \alpha + \sigma(\beta \times (r_{DB} - \mu))$, where $P^C_i = \frac{1}{C}\frac{1}{n_i}$ and $P^I = \frac{1}{C}\sum_{y^k_i=1}\frac{1}{n_i}$ are the resampling probabilities of instances in single-label and multi-label scenarios, respectively. This operation keeps instances with multiple labels from being oversampled, thus reducing redundant information about label co-occurrence. In summary, with the above improvements, the model can focus on learning labels that are hard to predict, i.e. tail-end codes in the ICD code space.

Algorithm 1 shows the pseudo-code of the training process.

Algorithm 1: The Training Process

$Data = \{x, y\}_{i=1}^{D}$: training dataset
Enc: text encoder for the representations of clinical document x and code synonyms L
Dec: decoder framework for multi-synonyms attention and multi-label classification
$T \leftarrow$ GenerateLabelTree (y)
$L \leftarrow$ SelectCodeSynonyms (y)
for k in range (T.maxLevel + 1) **do**
 for i in range ($Epochs[k]$) **do**
  for x, y in $Data$ **do**
  $H \leftarrow Enc(x)$
  $Q^L \leftarrow Enc(L)$
  $\hat{y} \leftarrow Dec_k(H, Q^L)$
  Loss $\leftarrow DBL(y, \hat{y})$

(continued on next column)

(continued)

Algorithm 1: The Training Process

  Loss.backward ()
  Optimizer.step ()
 end
end
if $k \neq T$.maxLevel **then**
 $Dec_{k+1}.Q \leftarrow Dec_k.Q$
else if $k = T$.maxLevel **then**
 $Dec_{k+1}.Q \leftarrow Dec_k.Q + fc(Q^L)$
end
end

4. Experiments

To verify the efficacy of our suggested model, this section conducts experiments and compares the results with baselines.

4.1. Datasets

MIMIC-III (Johnson et al., 2016): The Medical Information Mart for Intensive Care (MIMIC-III) is a widely used open-access medical database that contains over 50,000 patient EMRs stored in 26 tables. In the current study, we use the discharge summaries in the *noteevents* table as the input text and the corresponding ICD-9 codes in the *diagnoses_icd* and *procedure_icd* tables as the prediction label. Following the previous studies, we adopt two settings: MIMIC-III full and MIMIC-III 50, which contain discharge summaries with all 8,921 codes and the top 50 codes, respectively. Table 3 contains the statistics for the two datasets.

Pre-processing: Discharge summaries are textual documents that conclude all information about patients' hospitalisation, including past medical history, admission diagnosis, treatment records and discharge instructions. For fairness of comparison, we use the same pre-processing strategies as previous studies. First, the clinical texts are tokenized. Subsequently, we lowercase all tokens, swap out low-frequency tokens for 'UNK' and remove the non-alphabetic tokens. Based on all pre-processed texts, we use the Word2Vec CBOW method (Mikolov et al., 2013) to pre-train word embeddings, which will be used to encode code synonyms. Additionally, we truncated the tail of 311 documents that were longer than the limit of 4,096 tokens that Longformer can handle.

4.2. Evaluation metrics and baselines

Evaluation metrics: Following previous work, we use the same metrics to evaluate our methods, including macro-/micro-F1 and AUC (area under the ROC curve) and precision @K (P@K). 'Macro-average value' is the simple average of label-wise metrics, whereas 'micro-averaged value' is determined by considering each (text, code) pair as an individual prediction. Therefore, macro-average values can better reflect the performance of rare labels. 'P@K' indicates the proportion of the real labels in the top-K predicted labels, which we set $K \in \{8, 15\}$ for MIMIC-III-full and $K \in \{5, 8\}$ for MIMIC-III-50.

Baselines: The performance of our model is compared to previous models mentioned in related studies, including CNN-based models (CAML, DR-CAML, MultiResCNN, Hypercore, JAN), RNN-based models (LAAT, JointLAAT, HLAN, MSMN) and Transformer-based models (TransICD, BERT-ICD, Longformer-DLAC).

Table 3
Statistics of the MIMIC-III dataset.

	Train	Valid	Test	Train	Dev	Test
	MIMIC-III-full			MIMIC-III-50		
Documents	47,723	1631	3372	8066	1573	1729
#Ave tokens	1434	1724	1731	1478	1739	1763
#Ave codes	15.7	18.0	17.4	5.7	5.9	6.0

CAML and DR-CAML (Mullenbach et al., 2018**):** CAML uses CNNs as the text encoder and introduces a label-wise attention mechanism for ICD coding. DR-CAML further incorporates the ICD description to regularise the CAML model.

MultiResCNN (Li & Yu, 2020**):** Encoding texts with multi-filter residual CNNs, where a multi-filter convolutional layer is used to capture various text patterns and a residual block is adopted to enlarge the receptive field.

Hypercore (Cao et al., 2020a): Embedding ICD codes into the hyperbolic space to leverage code hierarchy and adopting GCNNs to capture code co-occurrence.

JAN (Wu et al., 2022**):** Combining document-based attention and label-based attention to leverage semantic information and presenting CorNet blocks to exploit code co-occurrence relations.

LAAT & JointLAAT (Vu et al., 2020**):** LAAT deploys Bi-LSTM and a label self-attention mechanism for automatic ICD coding. JointLAAT extends LAAT by applying hierarchical joint learning to capture the ICD code hierarchy.

HLAN (Dong et al., 2021**):** Using hierarchical label-wise attention networks to achieve word-level and sentence-level interpretations and capture the semantic relevance of codes through label embedding initialisation.

MSMN (Yuan et al., 2022**):** Using code synonyms and multiple synonyms matching networks to enrich ICD code representations and facilitate code prediction.

TransICD (Biswas et al., 2021**):** Introducing a label-wise attention layer into the Transformer architecture for ICD code prediction and using a label distribution aware margin (LDAM) loss function to address the label imbalance issue.

BERT-ICD (Pascual et al., 2021**):** Adopting 5 text truncation strategies to shorten clinical documents to 512 tokens and encoding with PubMedBERT.

Longformer-DLAC (Feucht et al., 2021**):** Deploying Longformer as the encoder and a description-based label attention classifier as the decoder to assign ICD codes.

4.3. Implementation details

Under the top 50 setting, we adopt Clinical-Longformer as the document encoder and predict the ICD codes using the proposed Hierarchical Curriculum Learning mechanism with Code Synonyms Attention and Distribution-Balanced Loss (HC-SA-DBL). Under the full setting, we focus on verifying the superiority of HC-SA-DBL by combining it with the encoder used in MultiResCNN (Li & Yu, 2020) and LAAT (Vu et al., 2020). We implement our model using Pytorch and the Hugging Face Transformers library (https://huggingface.co/yikuan8/Clinical-Longformer) on a 32 GB NVIDIA GeForce RTX 3090 Ti GPU. Referring to the previous studies, we summarise the detailed hyper-parameters in Table 4.

Table 4
Hyper-parameters for training MIMIC-III full and MIMIC-III 50.

Parameters	Full	Top 50
Input sequence length	4096	4096
word embedding dimesion (d_e)	100	100
hidden units size (d_h)	512, 300	768
Synonyms count (M)	4	8
DB loss parameters	$\alpha = 0.1$, $\beta = 10$, $\mu = 0.05$ and $\lambda = 2$	$\alpha = 0.1$, $\beta = 10$, $\mu = 0.05$ and $\lambda = 2$.
Optimizer	Adam	Adam
Batch size	8	4
Learning rate	5e-5	5e-5
Weight decay	0.1	0.1
Dropout rate	0.1	0.1
Epochs for five training rounds	2,3,5,10,50	2,3,5,10,50
Early stopping metric	P@8	P@5
Early stopping patience	10	10

The d_h of the Longformer encoder is set to 768 for the top 50 dataset. For the full dataset, we set the same d_h as MultiResCNN (Li & Yu, 2020) and LAAT (Vu et al., 2020), which encode the discharge summaries using Bi-LSTM and ResCNN, respectively. Regarding the synonym count, we perform a comparison over $M \in \{1, 2, 4, 8, 16\}$, resulting in the optimal value of $M = 8$ on the top 50 dataset (see Fig. 4). This result is consistent with the fact that the median and mean counts of UMLS synonyms are 5.0 and 5.4, respectively. Considering computational complexity, we adopt the suboptimal value $M = 4$ for the full dataset. In hierarchical curriculum learning, the query vector for child codes is initialised with the parent codes; thus, five-round training over the five-level code tree is needed. It should be noted that when constructing the code tree following (Ren et al., 2022), padding nodes in a level with few splits will engender a large number of redundant intermediate nodes. Thus, this leads to unnecessary computational burdens and possible overfitting of these codes. To alleviate this issue, we set epochs for five training rounds at 2, 3, 5, 10 and 50, respectively, where the layers with fewer splits are trained with fewer epochs.

4.4. Compared to SOTA methods

We repeat training 10 times with the same hyper-parameters using different random seeds for initialisation. The mean and deviation of the testing results over 10 runs are reported in Table 5 and Table 6.

Under the top 50 settings (Table 5), our model adopts Clinical-Longformer as the encoder and the proposed HC-SA-DBL as the decoder. Among the Transformer-based models, the BERT-ICD, which uses PubMedBERT to encode clinical documents, achieves better performance than Longformer-DLAC, which uses Longformer as the encoder to process 4,096 tokens. This result demonstrates the advantages of Longformer in modelling long clinical documents. The performance improvement of TransICD trained with LDAM loss illustrates the importance of balanced loss function. In our model (Clinical-Longformer + HC-SA-DBL), the encoder Clinical-Longformer has been pretrained using approximately 2 million clinical records extracted from the MIMIC-III dataset. It not only implies the medical knowledge of PubMedBERT but also inherits the advantage of Longformer in handling long text sequences, thus generating semantically rich clinical document representation with less information loss. Moreover, our model incorporates code hierarchy structure, code synonym embedding and DBL to help classify rare labels. For these reasons, our model scored higher on all metrics than the previous Transformer-based models.

Furthermore, our model achieved superior performance than CNN/RNN-based models on most metrics, except for the micro-AUC of MSMN. I suppose it is attributed to the fact that although Clinical-Longformer can handle 4,096 tokens, it still misses some crucial information at the end of clinical documents $> 4{,}096$, such as the discharge diagnosis. Regardless, the excellent ability of biomedical-specific PLMs to encode medical texts is evident to all. We believe that exploring more model combinations and parameter optimisation methods will help improve the performance of Transformer-based models in automatic ICD coding tasks.

Under the full setting, using Clinical-Longformer as the encoder is limited due to the demand for high computational resources. However, our suggested decoding architecture (HC-SA-DBL) can yet be combined with encoders from other studies to demonstrate its efficacy. Among CNN-based studies, conventional CNNs are usually used as encoders except for MultiResCNN, which designs multi-filter residual CNNs to encode clinical text and achieves superior performance. Among RNN-based studies, the Bi-LSTM encoder used in LAAT, JointLAAT and MSMN has demonstrated exceptional encoding capabilities. Moreover, MultiResCNN and LAAT use a simple label-wise attention mechanism as the decoder, which is suitable for comparison with our proposed HC-SA-DBL architecture. Therefore, under the full setting (Table 6), we focus on

Table 5
Results on MIMIC-III-50.

Methods	Model	AUC		F1		P@K	
		Macro	Micro	Macro	Micro	P@5	P@8
CNN	CAML	87.5	90.9	53.2	61.4	60.9	–
	DR-CAML	88.4	91.6	57.6	63.3	61.8	–
	HyperCore	**89.5**	92.9	60.9	66.3	63.2	–
	MultiResCNN	89.9	92.8	60.6	67.0	64.1	–
	JAN	89.8	**93.1**	**61.2**	**67.1**	**68.2**	
RNN	LAAT	92.5	94.6	66.6	71.5	67.5	54.7
	JointLAAT	92.5	94.6	66.1	71.6	67.1	54.6
	HLAN	88.4	91.9	57.1	64.1	62.5	–
	MSMN	**92.8**	**94.7**	**68.3**	**72.5**	**68.0**	–
Transformer	BERT-ICD	84.5	88.7	–	–	–	–
	Longformer-DLAC	87.0	91.0	52.0	62.0	61.0	–
	TransICD	89.4	92.3	56.2	64.4	61.7	–
	Clinical-Longformer + HC-SA-DBL (Ours)	**92.8**	**94.5**	**68.8**	**72.7**	**68.2**	**55.1**
		±0.09	±0.06	±0.32	±0.31	±0.26	±0.20

Table 6
Results on MIMIC-III full.

Methods	Model	AUC		F1		P@K	
		Macro	Micro	Macro	Micro	P@8	P@15
CNN	CAML	89.5	98.6	8.8	53.9	70.9	56.1
	DR-CAML	89.7	98.5	8.6	52.9	69.0	54.8
	MultiResCNN	91.0	98.6	8.5	55.2	73.4	58.4
	HyperCore	93.0	98.9	9.0	55.1	72.2	57.9
	JAN	**94.5**	**98.9**	**9.1**	**55.3**	**73.5**	**58.4**
RNN	LAAT	91.9	98.8	9.9	57.5	73.8	59.1
	JointLAAT	92.1	98.8	**10.7**	57.5	73.5	59.0
	MSMN	**95.0**	**99.2**	10.3	**58.4**	**75.2**	**59.9**
Ours	**MultiResCNN + HC-SA-DBL**	94.5	98.9	**11.6**	57.7	**75.4**	**59.9**
		±0.15	±0.04	±0.33	±0.41	±0.15	±0.17
	LAAT + HC-SA-DBL	**95.1**	**99.2**	11.4	**58.6**	74.7	59.8
		±0.09	±0.05	±0.31	±0.17	±0.20	±0.09

verifying the superiority of HC-SA-DBL by performing it after the encoder used in MultiResCNN and LAAT. HC leverages inheritance relationships between codes for model parameter migration. SA exploits the semantic information of code synonyms to enrich the code representation. DBL guides the model to focus on infrequency labels and considers code co-occurrence in re-weighting samples.

As shown in Table 6, HC-SA-DBL improves the corresponding baseline models (i.e. MultiResCNN and LAAT). For example, compared to MultiResCNN, the MultiResCNN + HC-SA-DBL has significant improvements in macro-AUC, micro-AUC, macro-F1, micro-F1, P@8 and P@15 by 3.5 %, 0.3 %, 3.1 %, 2.5 %, 2.0 % and 1.5 %, respectively. At the same time, LAAT + HC-SA-DBL is 3.2 %, 0.4 %, 1.5 %, 1.1 %, 0.9 % and 0.7 % higher than LAAT in these indicators. We note that the improvement in macro metrics is more significant than that in micro-metrics. That is because the MIMIC-III full dataset contains numerous rare codes whose performance is more relevant for macro metrics. In addition, our models achieved better or similar performance than CNN/RNN-based SOTA architectures. For example, compared to the best CNN-based model (i.e. JAN) that ignores the code hierarchy, our model scores higher on all metrics. Meanwhile, our model produces higher macro-AUC, F1 and P@8 than MSMN, which only exploits the code synonym representation. The above results reveal the efficiency of the presented decoder, which can considerably enhance the prediction performance of infrequent labels by taking code semantics, hierarchy and co-occurrence into account.

4.5. Ablation study

To further comprehend the contributions of each element in our model, we conducted an ablation study on the MIMIC-III 50 dataset. As shown in Table 7, we can find that all components help to increase the performance.

Table 7
Ablation study results on MIMIC-III-50.

Model	AUC		F1		P@K	
	Macro	Micro	Macro	Micro	P@5	P@8
Clinical-Longformer + HC-SA-DBL (ours)	**92.8**	**94.5**	**68.8**	**72.7**	**68.2**	**55.1**
w/o distribution-balance loss	91.4	94.0	64.8	70.5	67.4	54.7
w/o code synonym attention	91.2	93.6	64.6	70.1	67.2	54.6
w/o hierarchical curriculum learning	91.9	94.1	65.5	70.6	67.7	54.8
w/o Clinical-Longformer	92.1	94.2	66.4	70.9	67.8	54.9

(1) **Effectiveness of distribution-balance loss.** When the distribution-balance loss is replaced with the FL commonly used in previous studies (w/o distribution-balance loss), the macro-AUC, micro-AUC, macro-F1, macro-F1, P@5 and P@8 fall by 1.4 %, 0.5 %, 4.0 %, 2.2 %, 0.8 % and 0.2 %, respectively. These results demonstrate the importance of the code co-occurrence pattern, which is used to rebalance the sample weights to improve performance on rare codes.

(2) **Effectiveness of multi-synonym attention.** In this setting (w/o code synonym attention), we degrade multi-synonym attention to traditional label-wise attention and still use the hierarchical curriculum learning mechanism. Specifically, parameters from the parent node are used as the initial query vector and then modified by the code representation instead of synonym

representations. We find that this setting produces worse results on all indicators, which implies that the code synonym descriptions can facilitate better predictions by capturing richer semantic information about codes.

(3) **Effectiveness of hierarchical curriculum learning.** In the multi-synonym attention module, when we use only the code synonym representations as the query vector without hierarchical course learning (w/o hierarchical curriculum learning), the scores of all metrics have dropped. It indicates the effectiveness of knowledge transfer from upper-level nodes in the code hierarchy structure.

(4) **Effectiveness of the Clinical-Longformer.** The robustness of Word2Vec in the presence of noise and without the need for truncation was highlighted in (Corizzo et al., 2023; De Martino et al., 2023). Therefore, in the experiment (w/o Clinical-Longformer), we only used Word2Vec and the mean aggregation strategy to encode clinical documents, which led to worse results. This indicates that the richer medical knowledge in Clinical-Longformer is beneficial for extracting semantic representations of clinical texts. Moreover, Clinical-Longformer can generate dynamic word embeddings, outperforming Word2Vec in capturing global information and the polysemy of words in different contexts.

4.6. Visualisation

In the healthcare area, interpreting the predictive results is critical. Using the multi-synonym attention module, our model can assign various weights to tokens based on specific code synonyms, thus providing explainability. For each ICD code, we take the average of attention weights corresponding to all synonyms as the attention weights between this code and tokens. Several examples of visualising attention weights are shown in Fig. 2, which highlights the relevant tokens in the discharge summary for each code.

For the ICD-9 diagnosis code 401.9 (unspecified essential hypertension), our model successfully puts high weights on the keywords 'coumadin' and 'hypertension'. For the ICD-9 procedure code of 96.71 (continuous invasive mechanical ventilation for < 96 consecutive hours), the closely related words (i.e. 'lobectomy', 'intubated' and 'brief hospital') are highlighted. The visualisation outcomes demonstrate a strong correlation between our model's focus and human attention when conducting ICD coding. Therefore, this model would help professional coders improve coding efficiency and reliability. In addition, for incorrect code prediction, the explanation through highlighted keywords can provide a valuable reference for engineers to understand problems and improve the system.

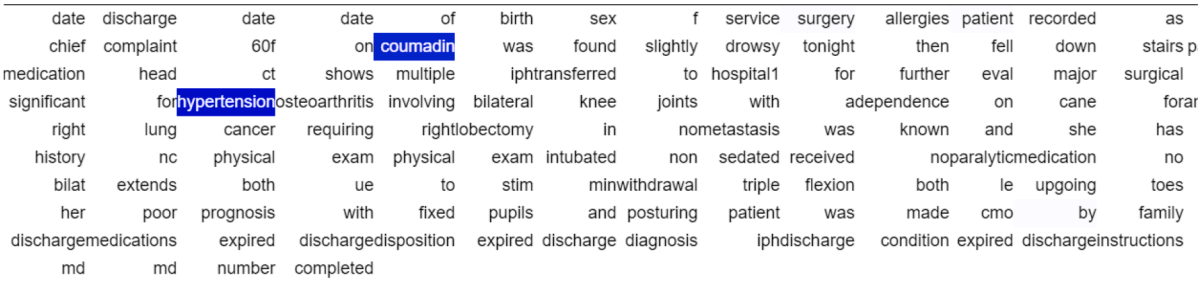
5. Discussion

5.1. Effect of pre-trained language models

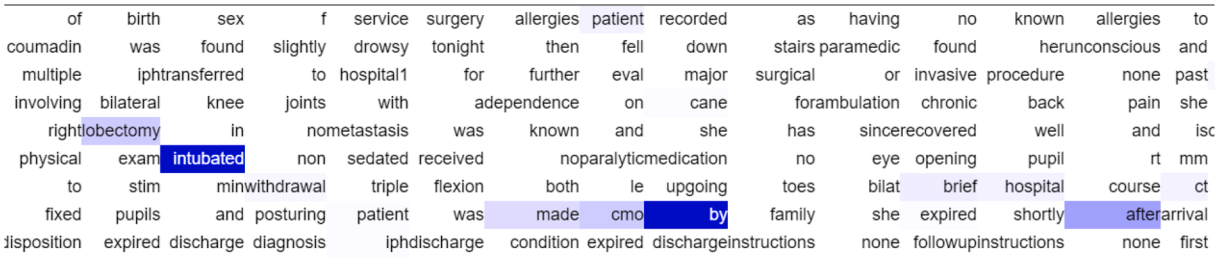
We experiment with several different PLMs as encoders and compare their performance with Clinical-Longformer in Table 8. BioBERT, PubMedBERT and ClinicalBERT are all BERT-based PLMs that are pre-trained on PubMed abstracts, PubMed paper full-text and MIMIC-III clinical notes, respectively. ClinicalXLNet is a pre-trained XLNet model that adopts MIMIC-III discharge summaries as the training corpus. XLNet is a generalised autoregressive PLM that overcomes the discrepancy problem brought by symbols (e.g. [MASK]) in the BERT model. Thus, XLNet-based PLMs outperformed BERT-based PLMs. As we all know, BERT limits the input sequence to 512 tokens. Although XLNet

Table 8
Results of different PLMs on MIMIC-III-50.

Model	AUC		F1		P@K	
	Macro	Micro	Macro	Micro	P@5	P@8
HC-SA-DBL + BioBERT	87.4	90.5	53.1	59.3	58.2	47.3
HC-SA-DBL + ClinicalBERT	87.3	90.7	55.8	60.9	59.7	47.5
HC-SA-DBL + PubMedBERT	90.5	92.0	59.1	65.5	63.2	52.6
HC-SA-DBL + ClinicalXLNet	91.5	93.8	64.5	70.0	66.3	54.6
HC-SA-DBL + Clinical-Longformer	**92.8**	**94.5**	**68.8**	**72.7**	**68.2**	**55.1**



(a) Code 401.9



(b) Code 96.71

Fig. 2. Attention visualisation examples.

allows for longer input sequences, the memory usage and computational complexity of XLNet grow quadratically as the input sequence expands. Thus, both BERT and XLNet face challenges in dealing with long clinical documents. So, we only use the first 512 tokens of discharge summaries as input in the above experiments. In our model, the Clinical-Longformer is based on the Longformer and is further trained on all MIMIC-III clinical notes. It can not only model long clinical documents up to 4,096 tokens but also embed knowledge from the MIMIC-III dataset, thus helping our model achieve the best performance.

5.2. Analysis of code correlation

To understand how code synonym descriptions and code structure help generate code representation, we use t-distributed stochastic neighbour embedding (t-SNE) to visualise the representation vector () of ICD codes. Fig. 3 shows the visualisation of the top 50 codes more clearly. It can be seen that our model perfectly separates the operation code (integer bit is 2) and the disease diagnosis code (integer bit is 3). At the same time, sibling codes (codes with the same parent) are clustered. For instance, the procedure codes 37.23 (partial excision of spleen) and 37.22 (total excision of spleen) are close to each other, whereas the adjacent diagnostic codes 272.0 (pure hypercholesterolemia) and 272.4 (other and unspecified hyperlipidaemia) have similar representations. The t-SNE results demonstrate that our model generates excellent code representations by successfully capturing the hierarchical structure and semantic relevance of ICD codes, which is beneficial for ICD coding tasks.

5.3. Complexity analysis

The computational complexity of our model mainly comes from two parts: the encoder and the decoder. The encoder is a pre-trained Clinical-Longformer, whose computational complexity stems from the updating of certain weights during the fine-tuning process. The decoder adds multi-head computation, hierarchical learning and DBL based on the classical label-wise attention mechanism. With fixed training data and parameters, its computational complexity is proportional to the number of heads (M) and code hierarchies. In the present study, we construct a five-level code tree based on the classification criteria of ICD codes, resulting in a five-round training over the tree structure. In addition, the setting of M depends on the number of code synonyms. After exploring the effect of different M on the model performance (see Fig. 4), we adopted the optimal setting $M = 8$, which led to an 8-fold increase in computational complexity during the last round of training. Compared to the best RNN-based model (e.g. MSMN), our decoder only adds four-round training over the first four levels of the code tree, which are set

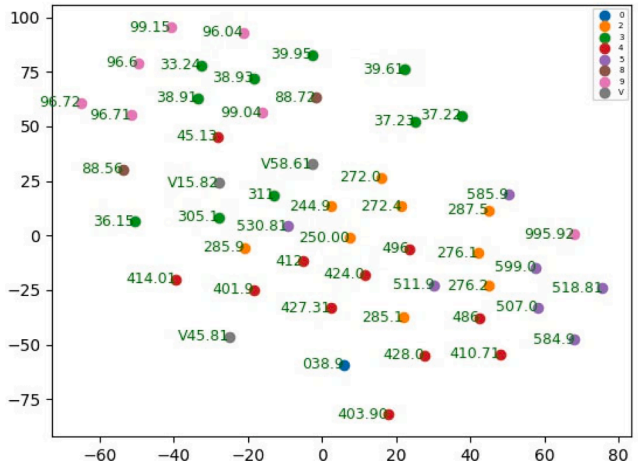


Fig. 3. T-SNE visualisation code representation.

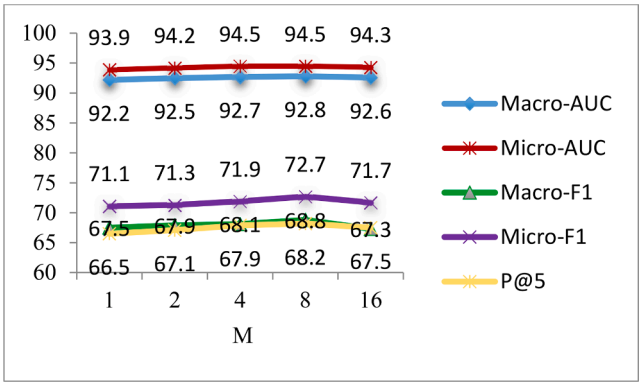


Fig. 4. Performance on MIMIC-III-50 at different synonym counts.

with fewer training epochs due to fewer nodes. Meanwhile, our model initialises the parameters of the child nodes using the trained parameters of the parent nodes. The knowledge accumulation in the first four levels promoted the faster convergence of our model than MSMN in the final-round training, as the parameters of MSMN are randomly initialised. Therefore, although our model requires five rounds of training, the total number of training epochs may be less than MSMN, which indicates the scalability of our decoder. In summary, as we understand, the introduction of ICD code synonyms and hierarchical structure improves the performance of code prediction at the expense of acceptable computing complexity.

6. Conclusion

Quickly and accurately assigning ICD codes to MERs is critical for billing, reimbursement and medical research. However, traditional manual coding is labour-intensive and error-prone. Therefore, many studies have been conducted for automatic ICD coding, which is usually modelled as a multi-label text classification problem and faces three main challenges: (1) the interpretability of coding; (2) lengthy clinical document and (3) long-tail label distribution.

In the present study, we propose a novel automatic ICD coding framework that can address the above issues by introducing the attention mechanism, biomedical-specific PLMs and latent code associations. First, we use the Clinical-Longformer to encode long clinical documents. The Clinical-Longformer has been continuously trained by Longformer using all MIMIC-III clinical notes. Therefore, it not only injects rich medical knowledge but also can model long text sequences up to 4,096 tokens, thus helping our model generate meaningful clinical document representation with less information loss. Moreover, we design a decoding architecture (namely HC-SA-DBL), which combines a multi-synonym attention mechanism, hierarchical curriculum learning and distribution-balance loss to perform ICD code prediction. The multi-synonym attention mechanism not only provides coding interpretation through attention weights but also enhances the code semantic representation using code synonym descriptions. The hierarchical curriculum learning mechanism enables knowledge transfer by leveraging the code hierarchy structure. The distribution-balance loss guides the model to focus on infrequency codes by re-weighting samples where code co-occurrence is considered. By jointly exploiting code synonym descriptions, code hierarchy and code co-occurrence, our model sufficiently captures code association, thereby improving the performance of tail-end codes. Experiments on a benchmark dataset (MIMIC-III) indicate that our proposed model yields competitive results compared to previous SOTA methods, especially improving the performance for infrequent codes.

The key contribution of this study is to present a novel and effective automatic ICD coding framework. Our encoded results by Clinical-Longformer can serve as a reference for future PLM-based automatic

ICD coding research. At the same time, our decoding architecture inspires other multi-label classification tasks with label association. Follow-up studies can further explore efficient document reduction methods to save the time of fine-tuning Transformers on long documents. In addition, applying Transformer variants designed for large label spaces, such as X-Transformer and XR-Transformer, to ICD coding is also a future research direction.

CRediT authorship contribution statement

Wen-hui Hou: Conceptualization, Methodology, Data curation, Writing – original draft, Visualization. **Xiao-kang Wang:** Investigation, Methodology, Data curation, Validation, Visualization. **Ya-nan Wang:** Investigation, Validation, Data curation. **Jian-qiang Wang:** Supervision, Writing – review & editing, Funding acquisition. **Fei Xiao:** Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

Alsentzer, E., Murphy, J. R., Boag, W., Weng, W.-H., Jin, D., Naumann, T., & McDermott, M. (2019). Publicly available clinical BERT embeddings. In *Proceedings of the 2nd clinical natural language processing workshop* (pp. 72–78). Minneapolis, Minnesota, USA: Association for Computational Linguistics.

Atutxa, A., Pérez, A., & Casillas, A. (2017). Machine learning approaches on diagnostic term encoding with the ICD for clinical documentation. *IEEE journal of biomedical and health informatics, 22*, 1323–1329.

Baumel, T., Nassour-Kassis, J., Cohen, R., Elhadad, M., & Elhadad, N. (2018). Multi-label classification of patient notes a case study on ICD code assignment. In *Proceedings of the Workshops of the Thirty-Second AAAI Conference on Artificial Intelligence* (pp. 409–416).

Biswas, B., Pham, T.-H., & Zhang, P. (2021). Transicd: Transformer based code-wise attention model for explainable icd coding. In *Proceedings of the 19th International Conference on Artificial Intelligence in Medicine,* (pp. 469-478). Springer, Cham.

Blanco, A., Casillas, A., Perez, A., & de Ilarraza, A. D. (2019). Multi-label clinical document classification: Impact of label-density. *Expert Systems with Applications, 138*, Article 112835.

Bodenreider, O. (2004). The unified medical language system (UMLS): Integrating biomedical terminology. *Nucleic acids research, 32*, 267–270.

Cao, P., Chen, Y., Liu, K., Zhao, J., Liu, S., & Chong, W. (2020a). Hypercore: Hyperbolic and co-graph representation for automatic icd coding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 3105-3114). Online: Association for Computational Linguistics.

Cao, P., Yan, C., Fu, X., Chen, Y., Liu, K., Zhao, J., Liu, S., & Chong, W. (2020b). Clinical-coder: Assigning interpretable ICD-10 codes to Chinese clinical notes. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations* (pp. 294-301). Online: Association for Computational Linguistics.

Corizzo, R., Pio, G., Barracchia, E. P., Pellicani, A., Japkowicz, N., & Ceci, M. (2023). Huri: Hybrid user risk identification in social networks. *World Wide Web, 26*, 3409–3439.

De Martino, G., Pio, G., & Ceci, M. (2023). Multi-view overlapping clustering for the identification of the subject matter of legal judgments. *Information Sciences, 638*, Article 118956.

Dong, H., Falis, M., Whiteley, W., Alex, B., Matterson, J., Ji, S., Chen, J., & Wu, H. (2022). Automated clinical coding: What, why, and where we are? *NPJ digital medicine, 5*, 159.

Dong, H., Suárez-Paniagua, V., Whiteley, W., & Wu, H. (2021). Explainable automated coding of clinical notes using hierarchical label-wise attention networks and label embedding initialisation. *Journal of biomedical informatics, 116*, Article 103728.

Farkas, R., & Szarvas, G. (2008). Automatic construction of rule-based ICD-9-CM coding systems. *BMC bioinformatics, 9*, 1–9.

Feucht, M., Wu, Z., Althammer, S., & Tresp, V. (2021). Description-based label attention classifier for explainable ICD-9 classification. In *Proceedings of the 2021 EMNLP Workshop W-NUT: The Seventh Workshop on Noisy User-generated Text* (pp. 62–66). Online: Association for Computational Linguistics.

Gao, S., Alawad, M., Young, M. T., Gounley, J., Schaefferkoetter, N., Yoon, H. J., Wu, X.-C., Durbin, E. B., Doherty, J., & Stroup, A. (2021). Limitations of transformers on clinical text classification. *IEEE journal of biomedical and health informatics, 25*, 3596–3607.

Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., Naumann, T., Gao, J., & Poon, H. (2021). Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare (HEALTH), 3*, 1–23.

Hu, S., Teng, F., Huang, L., Yan, J., & Zhang, H. (2021). An explainable CNN approach for medical codes prediction from clinical text. *BMC Medical Informatics and Decision Making, 21*, 1–12.

Huang, C.-W., Tsai, S.-C., & Chen, Y.-N. (2022). PLM-ICD: automatic ICD coding with pretrained language models. *arXiv:2207.05289*. https://doi.org/10.48550/arXiv.2207.05289.

Huang, J., Osorio, C., & Sy, L. W. (2019). An empirical evaluation of deep learning for ICD-9 code assignment using MIMIC-III clinical notes. *Computer methods and programs in biomedicine, 177*, 141–153.

Huang, K., Singh, A., Chen, S., Moseley, E. T., Deng, C.-Y., George, N., & Lindvall, C. (2020). Clinical XLNet: modeling sequential clinical notes and predicting prolonged mechanical ventilation. In *Proceedings of the 3nd clinical natural language processing workshop* (pp. s 94–100). Minneapolis, Minnesota, USA: Association for Computational Linguistics.

Johnson, A. E., Pollard, T. J., Shen, L., Lehman, L.-W.-H., Feng, M., Ghassemi, M., Moody, B., Szolovits, P., Anthony Celi, L., & Mark, R. G. (2016). MIMIC-III, a freely accessible critical care database. *Scientific data, 3*, 1–9.

Kaur, R., Ginige, J. A., & Obst, O. (2022). AI-based ICD coding and classification approaches using discharge summaries: A systematic literature review. *Expert Systems with Applications, 118997*.

Kim, B.-H., & Ganapathi, V. (2021). Read, attend, and code: pushing the limits of medical codes prediction from clinical notes by machines. In *Proceedings of the 6th Machine Learning for Healthcare Conference,* (pp. 196-208).

Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics, 36*, 1234–1240.

Li, F., & Yu, H. (2020). ICD coding from clinical text using multi-filter residual convolutional neural network. In *proceedings of the AAAI conference on artificial intelligence* (pp. 8180-8187). https://doi.org/10.1609/aaai.

Li, Y., Wehbe, R. M., Ahmad, F. S., Wang, H., & Luo, Y. (2022). Clinical-longformer and clinical-bigbird: Transformers for long clinical sequences. *arXiv:2201.11838*. https://doi.org/10.48550/arXiv.2201.11838.

Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision* (pp. 2980-2988).

Liu, Y., Cheng, H., Klopfer, R., Gormley, M. R., & Schaaf, T. (2021). Effective convolutional attention network for multi-label clinical document classification. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 5941-5953). Online: Association for Computational Linguistics.

Mayya, V., Kamath, S., Krishnan, G. S., & Gangavarapu, T. (2021). Multi-channel, convolutional attention based neural model for automated diagnostic coding of unstructured patient discharge summaries. *Future Generation Computer Systems, 118*, 374–391.

Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. In *Proceedings of 1st International Conference on Learning Representations (ICLR 2013)* (pp. 3111–3119).

Moradi, M., & Samwald, M. (2021). Explaining black-box models for biomedical text classification. *IEEE journal of biomedical and health informatics, 25*, 3112–3120.

Mullenbach, J., Wiegreffe, S., Duke, J., Sun, J., & Eisenstein, J. (2018). In *Explainable prediction of medical codes from clinical text* (pp. 1101–1111). Online: Association for Computational Linguistics.

Pascual, D., Luck, S., & Wattenhofer, R. (2021). In *Towards BERT-based automatic ICD coding: Limitations and opportunities* (pp. 54–63). Online: Association for Computational Linguistics.

Ren, W., Zeng, R., Wu, T., Zhu, T., & Krishnan, R. G. (2022). HiCu: Leveraging Hierarchy for Curriculum Learning in Automated ICD Coding. *In Proceedings of the 7th Machine Learning for Healthcare Conference (pp198–223)*.

Ridnik, T., Ben-Baruch, E., Zamir, N., Noy, A., Friedman, I., Protter, M., & Zelnik-Manor, L. (2021). Asymmetric loss for multi-label classification. In *In Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 82–91).

Rios, A., & Kavuluru, R. (2018). In *Few-shot and zero-shot multi-label learning for structured label spaces* (pp. 3132–3142). Brussels, Belgium: Association for Computational Linguistics.

Shi, H., Xie, P., Hu, Z., Zhang, M., & Xing, E. P. (2017). Towards automated ICD coding using deep learning. *arXiv:1711.04075*. https://doi.org/10.48550/arXiv.1711.04075.

Song, C., Zhang, S., Sadoughi, N., Xie, P., & Xing, E. (2021). Generalized zero-shot text classification for ICD coding. In *Proceedings of the twenty-ninth international joint*

conference on artificial intelligence (pp. 4018–4024). International Joint Conferences on Artificial Intelligence Organization.

Sun, W., Ji, S., Cambria, E., & Marttinen, P. (2022). Multitask balanced and recalibrated network for medical code prediction. *ACM Transactions on Intelligent Systems and Technology, 14*, 1–20.

Teng, F., Liu, Y., Li, T., Zhang, Y., Li, S., & Zhao, Y. (2022). A review on deep neural networks for ICD coding. *IEEE Transactions on Knowledge and Data Engineering, 35*, 4357–4375.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In Proceedings of the 31st Conference on Neural Information (pp. 5998–6000). CA, USA.

Vu, T., Nguyen, D. Q., & Nguyen, A. (2020). A label attention model for icd coding from clinical text. In *Proceedings of the twenty-ninth international joint conference on artificial intelligence* (pp. 3335–3341). International Joint Conferences on Artificial Intelligence Organization.

Wang, S., McDermott, M. B., Chauhan, G., Ghassemi, M., Hughes, M. C., & Naumann, T. (2020). In *Mimic-extract: A data extraction, preprocessing, and representation pipeline for mimic-iii* (pp. 222–235). New York, NY: Association for Computing Machinery.

Wu, T., Huang, Q., Liu, Z., Wang, Y., & Lin, D. (2020). Distribution-balanced loss for multi-label classification in long-tailed datasets. In Computer Vision – ECCV 2020. Lecture Notes in Computer Science (pp. 162-178). Glasgow, UK: Springer.

Wu, Y., Chen, Z., Yao, X., Chen, X., Zhou, Z., & Xue, J. (2022). JAN: Joint Attention Networks for Automatic ICD Coding. *IEEE journal of biomedical and health informatics, 26*, 5235–5246.

Xu, K., Lam, M., Pang, J., Gao, X., Band, C., Mathur, P., Papay, F., Khanna, A. K., Cywinski, J. B., & Maheshwari, K. (2019). Multimodal machine learning for automated ICD coding. In *Proceedings of machine learning research: vol. 106, Proceedings of the 4th machine learning for healthcare conference* (pp. 197–215). Ann Arbor, Michigan: PMLR.

Yu, Y., Li, M., Liu, L., Fei, Z., Wu, F.-X., & Wang, J. (2019). Automatic ICD code assignment of Chinese clinical notes based on multilayer attention BiRNN. *Journal of biomedical informatics, 91*, Article 103114.

Yuan, Z., Tan, C., & Huang, S. (2022). In *Code synonyms do matter: multiple synonyms matching network for automatic ICD coding* (pp. 808–814). Online: Association for Computational Linguistics.

Zhang, X., Xu, J., Soh, C., & Chen, L. (2022). La-hcn: Label-based attention for hierarchical multi-label text classification neural network. *Expert Systems with Applications, 187*, Article 115922.